

IBM Quantum vs. Classical Supercomputing: A Comparative Analysis of Processing Architectures for Complex Data Workloads

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ABSTRACT

The exponentially growing volume of data and the natural constraints of classical computing architectures bring serious challenges to complex optimization and molecular simulation work. This paper will review IBM Quantum, one of the most popular quantum computing platforms which operates on the concept of superconducting transmon qubits and contrast it with the classical supercomputing systems. Through processing model analysis, algorithmic complexity analysis and real-world implementation analysis, this paper shows that although IBM Quantum has the potential to achieve unprecedented parallel processing based on the principles of superposition and entanglement, it is currently limited by hardware noise and decoherence. A comparative study shows that classical systems continue to be better in deterministic and general-purpose workloads, but IBM Quantum offers a disruptive edge for certain highly complex computational problems. The paper concludes that as quantum error correction matures, IBM Quantum will have a paradigm shift in high-performance data processing.

Keywords: IBM Quantum, Superconducting Qubits, Quantum Advantage, High-Performance Computing, NISQ Architecture.

1.0 INTRODUCTION

Modern enterprises depend greatly on the high-performance data processing to extract insights out of the growing huge datasets. Historically, the scaling of classical supercomputers and distributed frameworks has been used to fulfill this need. Nonetheless, classical computing is limited by physical and architectural constraints especially when dealing with problems with exponentially large state spaces such as in quantum chemistry, cryptography, and optimization of complicated supply chains. The difficulty lies in the fact that

classical bits can only process information in a linear fashion and making the simulation of complicated quantum states is computationally infeasible. This paper will examine one of the leaders in cloud-based quantum computing, IBM Quantum and compares its processing power to the capabilities of classical supercomputing clusters. The objective of this paper is to explore the underlying architecture of IBM Quantum and evaluate its performance against classical systems as well as discuss its real world implications. The paper is structured through the explanation of the IBM Quantum technology, followed by a comparative analysis and an overview of the current limitations and future trends at the end of the paper.

2.0 LITERATURE REVIEW

The shift from classical computing to quantum computing is a radical shift in the capabilities of processing. In recent literature, the idea of quantum advantage has been highlighted, the hypothetical point at which quantum simulators surpass classical supercomputers in certain data intensive tasks. Practical quantum advantage is becoming a possibility to use when it comes to complex physical simulations (Daley et al., 2022). In addition, classical frameworks such as Apache Spark and Hadoop, as quantum frameworks process probabilistic states and binary data is processed deterministically (Gill et al., 2022). This taxonomy underscores the need for hybrid architectures, as quantum platforms like IBM Quantum are not general purpose replacements but specialized accelerators for exponentially complex workloads.

3.0 TECHNOLOGY AND ARCHITECTURE OF IBM QUANTUM

IBM quantum utilizes superconducting transmon qubits instead of using the classical bits which are the binary bits of 1s and 0s. Qubits are unique since it can be represented as 0, 1, or a combination of both at the same time through quantum superposition and this gives quantum computers

their unique power that can encode much richer information. The architecture for the IBM quantum is based on a dilution refrigerator which heavily focuses on the deep cryogenic cooling aspect in which temperatures are kept close to 15 millikelvins to reduce thermal noise (Bravyi et al., 2022). Moreover, in addition to traditional techniques, the thermodynamics properties of such systems are also being studied, with IBM quantum systems even being described as quantum batteries to understand their special energy characteristics (Gemme et al., 2022). The system comprises three components, the Quantum Processing Unit (QPU), the room-temperature microwave electronics that control the qubits and the cloud-based software that presents an open-source software development kit (SDK) Qiskit from IBM.

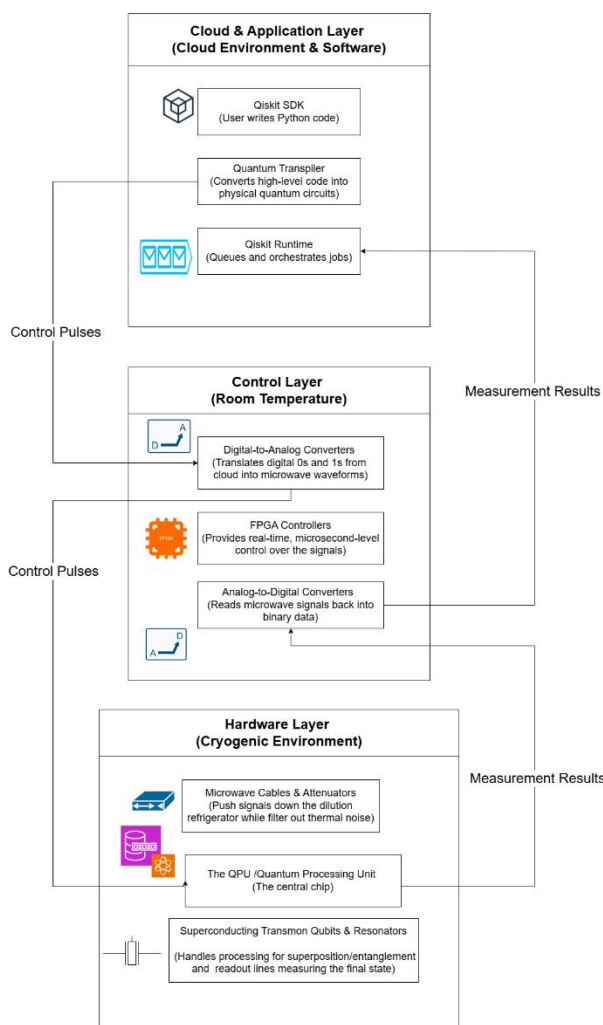


Figure 1. IBM Quantum System Architecture
(adapted from Gill et al., 2022)

Although it is powerful, the hardware is challenging to operate during the Noisy Intermediate-Scale Quantum (NISQ) era. Modern architectures are very vulnerable to decoherence and hardware error rates and need a large quantum volume to execute circuits with high reliability (Pelofske et al., 2022).

4.0 COMPARATIVE ANALYSIS AND REAL-WORLD APPLICATION

To understand the value of IBM Quantum, it is essential to compare it against the traditional high-performance classical supercomputing models such as those powered by advanced CPUs and GPUs.

Table 1. Comparison between Classical Supercomputing and IBM Quantum Computing

	CLASSICAL SUPERCOMPUTING	IBM QUANTUM COMPUTING
PROCESSING MODEL	Binary bits (1s and 0s); processed sequentially or parallel multi threads	Probabilistic qubits using superposition and entanglement.
SPEED AND SCALING	Linear scaling; struggles with exponential complexity	Exponential scaling for specific algorithms
FAULT TOLERANCE	Extremely high; negligible hardware error rates during data processing	Low; highly sensitive to decoherence and thermal noise.
MEMORY REQUIREMENT	Requires exponentially growing RAM to simulate larger complex systems.	Intrinsic state representation; N qubits represent 2^N states without added RAM.
BEST USE CASE SCENARIO	Large-scale batch processing, training deep neural networks, database queries.	Combinatorial optimization, cryptography, material science, and quantum chemistry.

Based on Table 1, while classical supercomputers are robust and fault-tolerant, they face hard physical limits when simulating nature or optimizing vast networks. IBM Quantum overcomes the memory and speed limitations for these specific domains, though it currently requires substantial error mitigation techniques due to the fragile nature of qubits. To benchmark high-performance data processing with IBM Quantum, it needs to be compared to conventional CPU/GPU supercomputing clusters. Demonstrations of the 127-qubit IBM device's ability to model physical systems that cannot be modelled by high-performance computing clusters show that it is useful even before full fault-tolerant error correction, which is still under development (Kim et al., 2023).

The architecture is being tested for commercial benefits. Commercial applications that require combinatorial optimization is essential. For instance, ExxonMobil uses IBM Quantum to simulate complex maritime transportation. ExxonMobil seeks to improve cargo transportation by mapping global variables of shipping to quantum circuits to minimize fuel costs and time in which a combinatorial problem that is computationally unmanageable by conventional high-performance computing clusters (Bova et al., 2021).

5.0 CONCLUSION

IBM Quantum is a significant breakthrough in the field of high-performance data processing as it has

moved the computability threshold. It can provide an exponential processing power over classical supercomputers by using superconducting qubits to solve problems with complex optimization and chemical reactions. Nevertheless, the comparative analysis shows that quantum computing is not a universal substitute to classical systems; but is a specialized accelerator. General workloads still require classical systems because current constraints such as high error rates and decoherence in the Noisy Intermediate-Scale Quantum (NISQ) era are too large to handle them (Pelofske et al., 2022). The next generation of quantum error correction research is to implement fault-tolerant quantum systems by using quantum error correction, enabling the development of hybrid quantum-classical data architecture (Bravyi et al., 2022).

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LOG BOOK

Date	Search Keywords	AI Prompts Used	Task Description
15 April 2026	IBM Quantum, Scopus, Web of Science, Quantum advantage	None	Searched for articles and research papers using Web of Science on IBM Quantum performance. Found 15 relevant articles.
18 April 2026	Quantum vs Classical supercomputing, ExxonMobil IBM	Explain the concept of quantum advantage in simpler terms to help me outline my literature review	Read only the abstracts and conclusions to filter the references and lists down to 7 most relevant WoS-indexed papers.
22 April 2026	IBM Quantum architecture diagram, Superconducting qubits	Review my comparison table criteria for classical vs. quantum computing. Are the metrics for fault tolerance accurate for the NISQ era?	Drafted the Literature Review and Architecture sections. Designed the architecture diagram in draw.io.
26 April 2026	APA 7th reference format, IEEE paper template	Check this section for academic tone and check that my APA 7th in-text citations are formatted correctly.	Finalized the comparative analysis table. Polished all text for academic tone. Formatted to IEEE standards and prepared Turnitin report.